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https://www.tensorflow.org/api_docs/python/tf/clip_by_norm

Clips tensor values to a maximum L2-norm.

Function Signatures

```
tf.clip_by_norm( t, clip_norm, axes=None, name=None )
```

Code Examples

```
tf.clip_by_norm(  
t, clip_norm, axes=None, name=None  
)
```

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```
some_nums = tf.constant([[1, 2, 3, 4, 5]], dtype=tf.float32)  
tf.clip_by_norm(some_nums, 2.0).numpy()  
array([[0.26967996, 0.5393599 , 0.80903983, 1.0787199 , 1.3483998  
]],  
dtype=float32)
```

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tf.clip_by_norm(some_nums, 2.0).numpy()
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array([[0.26967996, 0.5393599 , 0.80903983, 1.0787199 , 1.3483998  
]],
```

```
dtype=float32)
```

```
# Get your gradients after training  
loss_value, grads = grad(model, features, labels)  
# Apply some clipping  
grads = [tf.clip_by_norm(g, norm)  
for g in grads]  
# Continue on with training  
optimizer.apply_gradients(grads)
```

Documentation

Clips tensor values to a maximum L2-norm.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.clip_by_norm`

Used in the notebooks

- Advanced automatic differentiation

Given a tensor `t`, and a maximum clip value `clip_norm`, this operation normalizes `t` so that its L2-norm is less than or equal to `clip_norm`, along the dimensions given in `axes`. Specifically, in the default case where all dimensions are used for calculation, if the L2-norm of `t` is already less than or equal to `clip_norm`, then `t` is not modified. If the L2-norm is greater than `clip_norm`, then this operation returns a tensor of the same type and shape as `t` with its values set to:

$$t * clip_norm / l2norm(t)$$

In this case, the L2-norm of the output tensor is `clip_norm`.

As another example, if `t` is a matrix and `axes == [1]`, then each row of the output will have L2-norm less than or equal to `clip_norm`. If `axes == [0]` instead, each column of the output will be clipped.

Code example:

This operation is typically used to clip gradients before applying them with an optimizer. Most gradient data is a collection of different shaped tensors for different parts of the model. Thus, this is a common usage:

Returns

Raises
